Synoptic Question 30: The Physics of an Automatic Washing Machine

Reading Passage

Read the following article and then answer the questions that follow.

An automatic washing machine uses a combination of mechanical action, thermal energy, and chemical agents (detergents) to clean clothes. A key component is a rotating drum, usually mounted horizontally or slightly tilted. Clothes are placed inside the drum, which is then partially filled with water.

An electric motor drives the drum's rotation. During the wash cycle, the drum rotates back and forth at a relatively low speed, tumbling the clothes through the water. For the spin cycle, the drum rotates at a very high speed. The resulting centripetal force pushes the clothes against the perforated drum wall, while the water passes through the holes and is pumped out, effectively drying the clothes.

Many washing machines also include an electrical heating element submerged in the water. This element, essentially a large resistor, converts electrical energy into thermal energy to heat the water to a desired temperature, which improves the effectiveness of the detergent. The power consumption of the machine is dominated by the heating element and the motor, particularly during the high-speed spin cycle.

Modern machines often use electronic controls to manage the cycle timing, water level, temperature, and motor speed. The motor itself is typically an AC induction motor or a brushless DC motor, chosen for its efficiency and ability to provide variable speeds.

Data:

Data for a typical front-loading washing machine:

Table 1: Data for Washing Machine Analysis

Parameter	**Value**
Inner radius of drum, r	0.25 m
Length of drum, L	0.30 m
Mass of empty drum, m_{drum}	8.0 kg
Maximum mass of dry clothes, m_{load}	7.0 kg
Volume of water used in wash cycle	50 litres
Density of water, ρ_{water}	1000 kg m^{-3}
Specific heat capacity of water, c_{water}	4200 J kg^{-1} K^{-1}
Rotational speed during wash cycle	50 rpm
Rotational speed during spin cycle, $\omega_{spin,rpm}$	1200 rpm
Time taken to accelerate to spin speed, t_{accel}	4.0 s
Power rating of heating element, P_{heat}	2100 W
Mains voltage supply (RMS), V_{rms}	230 V
Efficiency of motor during spin, η_{motor}	75%

Questions

(a) During the wash cycle, water is heated.

(i) Calculate the mass of water used in the wash cycle. (1 litre = 1×10^{-3} m^3).

(ii) Calculate the energy required to heat this water from 15.0 °C to 40.0 °C.

(iii) Assuming the heating element is the only component drawing power, calculate the time taken to heat the water.

(iv) Calculate the resistance of the heating element.

(b) During the spin cycle, the drum rotates at high speed. Assume the clothes (m_{load}) stick uniformly to the inner wall of the drum.

(i) Convert the spin speed of 1200 rpm into angular velocity ω in rad s^{-1}.

(ii) Calculate the linear speed v of the clothes during the spin cycle.

(iii) Calculate the magnitude of the centripetal force acting on a small 0.10 kg piece of clothing during the spin.

(iv) Explain qualitatively why the water is extracted from the clothes during the spin cycle.

(c) Consider the acceleration phase of the spin cycle. The total moment of inertia I_{total} of the drum and wet clothes can be approximated as $I_{total} \approx (m_{drum} + m_{wet_load})r^2$. Assume the wet load mass is double the dry load mass.

(i) Estimate the moment of inertia I_{total} of the loaded drum.

(ii) Assuming constant angular acceleration, calculate the angular acceleration α as the drum speeds up.

(iii) Calculate the net torque τ_{net} required to produce this acceleration.

(iv) Calculate the work done W by this net torque during the acceleration phase.

(d) The motor provides the torque to accelerate and spin the drum.

(i) Calculate the maximum rotational kinetic energy of the loaded drum during the spin cycle.

(ii) Assuming the work done in (c)(iv) equals the final rotational KE, calculate the average mechanical power P_{mech} supplied by the motor during acceleration.

(iii) Calculate the average electrical power P_{elec} consumed by the motor during acceleration.

(e) Briefly explain the basic principle (interaction between magnetic fields and currents) by which an electric motor produces torque.

Solutions

(a) (i) Volume $V = 50$ litres $= 50 \times 10^{-3}$ m^3.
Mass $m_{water} = \rho_{water} V = (1000)(50 \times 10^{-3}) = 50.0$ kg.

(ii) $\Delta T = 40.0 - 15.0 = 25.0$ K.
Energy $Q = m_{water} c_{water} \Delta T = (50.0)(4200)(25.0) = 5.25 \times 10^6$ J.

(iii) Power $P_{heat} = 2100$ W.
Time $t = \frac{Q}{P_{heat}} = \frac{5.25 \times 10^6}{2100} = 2500$ s (approx 41.7 minutes).

(iv) $P = \frac{V_{rms}^2}{R} \implies R = \frac{V_{rms}^2}{P_{heat}}$
$R = \frac{(230)^2}{2100} = 25.19\ \Omega \approx 25.2\ \Omega$.

(b) (i) $\omega = 1200 \frac{\text{rev}}{\text{min}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} \times \frac{1 \text{ min}}{60 \text{ s}} = 125.7$ rad s^{-1}.

(ii) $v = r\omega = (0.25)(125.7) = 31.43$ m s$^{-1} \approx 31.4$ m s^{-1}.

(iii) $m_{piece} = 0.10$ kg.
$F_c = m_{piece} r \omega^2 = (0.10)(0.25)(125.7)^2 = 395$ N.
(Alternatively: $F_c = m_{piece} v^2 / r = (0.10)(31.43)^2/0.25 = 395$ N).

(iv) The drum wall provides the large inward centripetal force required to keep the clothes moving in a circle. Water droplets, being mobile, are not held as strongly by the fabric. Their inertia means they tend to continue moving in a straight line (tangential to the circle). They pass through the perforations in the drum and are collected and pumped away.

(c) $m_{wet_load} = 2 \times m_{load} = 2 \times 7.0 = 14.0$ kg.

(i) $I_{total} \approx (m_{drum} + m_{wet_load}) r^2 = (8.0 + 14.0)(0.25)^2 = (22.0)(0.0625) = 1.375$ kg m^2.

(ii) $\omega_f = 125.7$ rad s^{-1}, $\omega_i = 0$.
$\alpha = \frac{\omega_f - \omega_i}{t_{accel}} = \frac{125.7 - 0}{4.0} = 31.4$ rad s^{-2}.

(iii) $\tau_{net} = I_{total} \alpha = (1.375)(31.4) = 43.175$ Nm ≈ 43.2 Nm.

(iv) Angular displacement $\theta = \omega_i t + \frac{1}{2}\alpha t^2 = 0 + \frac{1}{2}(31.4)(4.0)^2 = 251.2$ rad.
Work done $W = \tau_{net}\theta = (43.175)(251.2) = 10845$ J $\approx 1.08 \times 10^4$ J.
(Alternatively: $W = \Delta KE_{rot} = \frac{1}{2} I \omega_f^2 = \frac{1}{2}(1.375)(125.7)^2 = 10857$ J). Let's use 10.9 kJ.

(d) (i) $KE_{rot,max} = \frac{1}{2} I_{total} \omega_{spin}^2 = \frac{1}{2}(1.375)(125.7)^2 = 10857$ J ≈ 10.9 kJ.

(ii) $P_{mech,avg} = \frac{W}{t_{accel}} = \frac{10857}{4.0} = 2714$ W ≈ 2.71 kW.

(iii) $\eta_{motor} = \frac{P_{mech}}{P_{elec}} \implies P_{elec} = \frac{P_{mech}}{\eta_{motor}}$
$P_{elec,avg} = \frac{2714}{0.75} = 3619$ W ≈ 3.62 kW.

(e) An electric motor typically uses coils carrying an electric current placed within a magnetic field. The magnetic field exerts a force (the Lorentz force, $F = BIL \sin\theta$) on the current-carrying wires. These forces create a turning effect, or torque, on the coil assembly (rotor), causing it to rotate and drive the drum.

Synoptic Question 30 (Variant 2 — Higher Difficulty)

Reading Passage

An advanced front-loading washing machine is controlled to minimise energy and water use while still achieving effective cleaning. During the wash phase, a heater raises water temperature. During the spin phase, the drum reaches high angular speed so that water is expelled through perforations.

For design and safety, engineers must model rotational dynamics, power conversion, transient acceleration, thermal losses, and structural loading. In this variant, assume the clothes form a rotating annulus near the drum wall during spin, and that the motor controller can apply regenerative braking during deceleration.

Data for Variant 2

Parameter	Value
Drum inner radius, r	0.26 m
Drum length, L	0.32 m
Empty drum mass, m_{drum}	8.5 kg
Dry clothes mass, m_{dry}	8.0 kg
Water used in wash cycle	42 L
Initial water temperature	12 °C
Target water temperature	55 °C
Specific heat capacity of water, c	4200 $\text{Jkg}^{-1}\text{K}^{-1}$
Heater power rating, P_h	2.20 kW
Mains supply (RMS), V_{rms}	230 V
Spin speed	1400 rpm
Time from rest to spin speed	5.0 s
Motor mechanical efficiency (motoring), η_m	0.82
Motor-generator efficiency (regen), η_r	0.60
Fraction of dry mass retained as water at start of spin	0.85
Deceleration time from spin speed to rest	6.0 s

Take density of water as $\rho = 1000\,\text{kgm}^{-3}$ and $g = 9.81\,\text{ms}^{-2}$.

Questions

(a) **Thermal-electrical analysis**

(i) Determine the mass of water heated in the wash cycle.

(ii) Calculate the ideal thermal energy needed to heat this water from 12 °C to 55 °C.

(iii) If measured heating time is 48 min, estimate the overall heating efficiency (useful thermal output / electrical input).

(iv) Determine the heater resistance at rated voltage.

(b) **Spin kinematics and force**

(i) Convert 1400 rpm to angular velocity in rads^{-1}.

(ii) Find the tangential speed at the drum wall.

(iii) Compute the centripetal acceleration in units of g.

(iv) A 0.12 kg fabric section is at the drum wall. Find the centripetal force on it.

(c) **Rotational dynamics and power flow**

Assume the wet load mass at spin start is

$$m_{\text{wet}} = m_{\text{dry}}\,(1 + 0.85)\,.$$

Model the drum + wet load as a thin cylindrical shell at radius r, so

$$I \approx (m_{\text{drum}} + m_{\text{wet}})\,r^2.$$

(i) Calculate I.

(ii) Calculate angular acceleration during spin-up.

(iii) Determine required net torque.

(iv) Determine rotational kinetic energy at full spin.

(v) Determine average mechanical power during spin-up.

(vi) Estimate average electrical input power to the motor during spin-up.

(d) **Regenerative braking estimate**

(i) Assuming all rotational KE at full speed is available to braking, find the ideal recoverable energy.

(ii) Using $\eta_r = 0.60$, estimate electrical energy returned to the supply/storage.

(iii) Compute average regenerated electrical power over 6.0 s deceleration.

(e) **Open-ended conceptual question (physics explanation)**

In up to 200 words, discuss the statement: *"Water leaves the clothes in the spin cycle because the centrifugal force throws it outward."*

Your answer should critically evaluate this statement using an inertial-frame explanation and may also comment on how the rotating-frame description is used in engineering practice.

(f) **Open-ended design question (extended response)**

A manufacturer wants to increase spin speed from 1400 rpm to 1600 rpm to reduce residual moisture, while keeping peak electrical input power unchanged.

Propose and justify **two** engineering modifications. Your response should include relevant physics principles (rotational dynamics, stress/vibration, motor control, fluid extraction, or energy management) and discuss one trade-off for each modification.

Sample Solutions (Variant 2)

(a) Thermal-electrical analysis

(i) $m = \rho V = (1000)(42 \times 10^{-3}) = 42.0\,\text{kg}$.

(ii) $\Delta T = 55 - 12 = 43\,\text{K}$,

$$Q = mc\Delta T = (42)(4200)(43) = 7.5852 \times 10^6\,\text{J} \approx 7.59\,\text{MJ}.$$

(iii) Electrical input over 48 min:

$$E_{\text{in}} = P_h t = (2200)(48 \times 60) = 6.336 \times 10^6\,\text{J}.$$

Heating efficiency

$$\eta_h = \frac{Q}{E_{\text{in}}} = \frac{7.5852 \times 10^6}{6.336 \times 10^6} = 1.20\ (120\%).$$

This is non-physical ($> 100\%$), indicating at least one assumption/data value is inconsistent (for example, real heated water mass smaller than 42 kg, heater not at rated power continuously in the measured interval, or data/measurement error).

(iv)

$$R = \frac{V_{\text{rms}}^2}{P_h} = \frac{230^2}{2200} = 24.0\,\Omega\ (3\,\text{s.f.}).$$

(b) Spin kinematics and force

(i)

$$\omega = 1400\frac{\text{rev}}{\text{min}} \cdot \frac{2\pi\,\text{rad}}{1\,\text{rev}} \cdot \frac{1\,\text{min}}{60\,\text{s}} = 146.6\,\text{rads}^{-1}.$$

(ii) $v = r\omega = (0.26)(146.6) = 38.1\,\text{ms}^{-1}$.

(iii)

$$a_c = r\omega^2 = (0.26)(146.6)^2 = 5.59 \times 10^3\,\text{ms}^{-2}, \qquad \frac{a_c}{g} = \frac{5.59 \times 10^3}{9.81} \approx 5.70 \times 10^2.$$

So the load experiences about $570g$.

(iv)

$$F_c = ma_c = (0.12)(5.59 \times 10^3) = 6.71 \times 10^2\,\text{N} \approx 671\,\text{N}.$$

(c) Rotational dynamics and power flow

Wet mass: $m_{\text{wet}} = 8.0(1 + 0.85) = 14.8\,\text{kg}$.
Total rotating mass: $m_T = 8.5 + 14.8 = 23.3\,\text{kg}$.

(i)

$$I \approx m_T r^2 = (23.3)(0.26)^2 = 1.58\,\text{kgm}^2.$$

(ii)

$$\alpha = \frac{\Delta\omega}{\Delta t} = \frac{146.6}{5.0} = 29.3\,\text{rads}^{-2}.$$

(iii)

$$\tau = I\alpha = (1.58)(29.3) = 46.2\,\text{Nm}.$$

(iv)

$$E_{k,\mathrm{rot}} = \frac{1}{2}I\omega^2 = \frac{1}{2}(1.58)(146.6)^2 = 1.70 \times 10^4\,\mathrm{J} \approx 17.0\,\mathrm{kJ}.$$

(v)

$$\bar{P}_{\mathrm{mech}} = \frac{\Delta E_k}{\Delta t} = \frac{1.70 \times 10^4}{5.0} = 3.40\,\mathrm{kW}.$$

(vi)

$$\bar{P}_{\mathrm{elec}} = \frac{\bar{P}_{\mathrm{mech}}}{\eta_m} = \frac{3.40}{0.82} = 4.15\,\mathrm{kW}.$$

(d) Regenerative braking estimate

(i) Ideal recoverable energy equals rotational KE at spin start:

$$E_{\mathrm{ideal}} = 17.0\,\mathrm{kJ}.$$

(ii) Returned electrical energy:

$$E_{\mathrm{return}} = \eta_r E_{\mathrm{ideal}} = 0.60(17.0\,\mathrm{kJ}) = 10.2\,\mathrm{kJ}.$$

(iii) Average regenerated power over 6.0 s:

$$\bar{P}_{\mathrm{regen}} = \frac{E_{\mathrm{return}}}{6.0} = 1.70\,\mathrm{kW}.$$

(e) Open-ended conceptual question — sample response

In an inertial frame, no real outward force is required. The drum wall supplies an inward contact force that constrains cloth parcels to circular motion. Water that is weakly bound in the fabric has less constraint, so when local contact is lost it tends to continue approximately tangentially (straight-line inertia), not follow the curved drum path. Relative to the rotating drum, this appears as outward migration toward perforations. So saying "centrifugal force throws water out" is incomplete in fundamental terms: in an inertial description, separation occurs because cloth is forced inward while less-constrained water is not. In a rotating frame fixed to the drum, introducing a centrifugal pseudo-force is mathematically useful for predicting pressure gradients and effective extraction, provided one states it is a non-inertial-frame construct rather than a real interaction force.

(f) Open-ended design question — sample response

Modification 1: Increase effective drum radius slightly while lowering target angular speed profile for equivalent rim speed where feasible. Since extraction correlates with high normal acceleration $a = r\omega^2$, a modest radius increase can help maintain extraction while reducing required ω rise rate, lowering peak acceleration torque and current demand. **Trade-off:** larger radius increases cabinet size, material cost, and potential bearing load.

Modification 2: Use model-based motor control with longer spin-up and active imbalance detection/correction. By extending ramp time, peak electrical power can be held constant while still reaching higher final speed. Improved balancing reduces vibration and parasitic losses, allowing more of the fixed electrical budget to reach useful rotational KE. **Trade-off:** longer cycle time and higher control-system complexity.

Mark Scheme (Variant 2)

Total: 50 marks

- **(a) Thermal-electrical analysis [10]**: (i) 2, (ii) 3, (iii) 3 for efficiency + interpretation of non-physical outcome, (iv) 2.
- **(b) Spin kinematics and force [10]**: (i) 2, (ii) 2, (iii) 3, (iv) 3.
- **(c) Rotational dynamics and power flow [12]**: (i) 2, (ii) 2, (iii) 2, (iv) 3, (v) 1, (vi) 2.
- **(d) Regenerative braking estimate [6]**: (i) 2, (ii) 2, (iii) 2.
- **(e) Open-ended conceptual response [6]**:
  - 1 mark: identifies inertial-frame perspective.
  - 2 marks: explains inward constraint force for clothes / circular motion requirement.
  - 2 marks: explains why less-constrained water separates and exits through perforations.
  - 1 mark: correctly distinguishes pseudo-force usage in rotating frame.
- **(f) Open-ended design response [6]**:
  - 2 marks: first modification physically valid and justified.
  - 2 marks: second modification physically valid and justified.
  - 2 marks: one clear trade-off for each modification.

Quality of working rule (applies globally): award method marks for correct physics setup even if arithmetic slips occur; full credit requires consistent units and appropriate significant figures.